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MODULE 1 — DATA WAREHOUSING FUNDAMENTALS

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THEORY QUESTIONS

- Compare OLTP vs OLAP
- Why every DW structure contains a time element
- Basic building blocks of a data warehouse
- ER modelling vs Dimensional modelling
- Difference between top-down and bottom-up DW approaches
- Dimensional modeling + star/snowflake schema for:
 - Furniture company
 - Social media interactions
- Updates to dimensional table
- Star vs Snowflake schema
- Fact constellation schema
- Techniques of data loading
- Major steps in ETL process

NUMERICALS

- Create cube + describe OLAP (Slice/Dice/Rollup/Drill/Pivot)

FREQUENCY

High frequency repeated topics:

1. OLTP vs OLAP — **6+ times**
2. Dimensional modeling + schema design — **7+ times**
3. ETL steps — **4+ times**
4. Time element in DW — **2 times**
5. OLAP cube numericals — **3 times**

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MODULE 2 — INTRO TO DATA MINING, EXPLORATION, PREPROCESSING

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THEORY QUESTIONS

- KDD process (with diagram)
- Issues in data mining
- Types of attributes
- Data visualization techniques
- Data preprocessing steps
- Handling missing values
- Applications of data mining
- Discretization & Concept hierarchy generation
- Dimensionality reduction, transformation
- Techniques to handle noisy data
- Data exploration basics

FREQUENCY

Dominant repeated topics:

1. KDD process — **7+ times**
2. Data preprocessing — **6+ times**
3. Issues in data mining — **5 times**
4. Data visualization — **4 times**
5. Missing values handling — **3 times**

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MODULE 3 — CLASSIFICATION

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THEORY QUESTIONS

- Methods for estimating classifier accuracy
- Holdout, random subsampling, cross-validation, bootstrap
- Issues in classification & prediction
- Decision tree induction algorithm
- Decision tree based classification with example
- Naive Bayes working + naive assumption + example
- Difference between classification and clustering
- Supervised vs unsupervised learning

NUMERICALS

- Naive Bayes on dataset
- ID3 decision tree build
- Classifier accuracy evaluation numericals

FREQUENCY

Most repeated:

1. Accuracy estimation methods — **7+ times**
2. Naive Bayes — **5+ times**
3. Decision tree (ID3) — **5+ times**
4. Classification vs clustering — **3 times**

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MODULE 4 — CLUSTERING

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THEORY QUESTIONS

- Types of clustering
- K-Means algorithm (+ flowchart)
- K-Medoids algorithm
- Agglomerative vs Divisive clustering
- Issues in clustering

NUMERICALS

- K-Means on given dataset (multiple papers)
- Single-link clustering + dendrogram
- Complete link / Agglomerative methods
- Clustering on 1-D and 2-D datasets

FREQUENCY

Repeated heavily:

1. K-Means — **8+ times**
2. Single-link dendrogram — **4 times**
3. Agglomerative vs divisive — **3 times**
4. K-Medoids — **2 times**

MODULE 5 — FREQUENT PATTERNS & ASSOCIATION RULES

THEORY QUESTIONS

- Market Basket Analysis
- Apriori algorithm
- Multilevel association rule mining
- Multidimensional association rule mining
- FP-Tree
- FP-Growth algorithm

NUMERICALS

- Apriori on datasets (support/confidence given)
- FP-Growth numerical

FREQUENCY

Heaviest repetition:

1. Apriori — **10+ times**

2. Multilevel association rules — **6+ times**
3. FP-Tree / FP-Growth — **5 times**
4. Market basket analysis — **4 times**

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MODULE 6 — WEB MINING

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THEORY QUESTIONS

- Web structure mining
- PageRank (with example)
- Web usage mining (repeated heavily)
- Web content mining
- CLARANS extension
- Web mining vs Data mining
- Types of web mining
- Applications of web mining

FREQUENCY

Most repeated by far in all papers:

1. PageRank — **8+ times**
2. Web usage mining — **7+ times**
3. Types of web mining — **6 times**
4. CLARANS — **2 times**
5. Web content mining — **3 times**

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FINAL CONSOLIDATED FREQUENCY SUMMARY

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Module	Most Repeated Topics	Frequency
1. DW Fundamentals	OLTP vs OLAP, Dimensional modelling, Schema design	Very High
2. Preprocessing + KDD	KDD, Preprocessing, Issues in DM	Very High
3. Classification	Accuracy methods, Naive Bayes, Decision tree	High
4. Clustering	K-Means, dendrogram	High
5. Association Rules	Apriori, Multilevel rules, FP-growth	Very High
6. Web Mining	PageRank, Web usage mining, Types of WM	Extremely High

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